Fewer Babies, Bigger Cities

AP Human Geography · Units 2 & 6 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

The table shows data for four countries. (The countries are fictional and the data were created for practice.)

Selected Population and Development Indicators

Country	Total fertility rate 1995	Total fertility rate 2020	Female secondary school enrollment, 2020 (%)	Urban population, 2020 (%)	Infant mortality, 2020 (per 1,000 live births)
Norvale	6.1	5.4	22	24	68
Kestria	5.8	3.1	61	48	31
Tamboa	3.4	1.9	88	71	12
Ruzhan	2.0	1.3	97	82	3

Total fertility rate = average number of children per woman. Data created for practice.

A. Identify the country in the table that is most likely in Stage 2 of the demographic transition model. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Norvale

MODEL ANSWER

Norvale.

Common mistake: "Ruzhan." Ruzhan has the LOWEST TFR (1.3). That fits Stage 4 or 5, not Stage 2.

Why the model works: Norvale has the highest TFR (5.4) and infant mortality and is the most rural: all Stage 2 signs.

B. Describe the relationship between female secondary school enrollment and total fertility rate shown in the table. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- States an inverse/negative relationship (more enrollment, lower TFR)
- Supports it with numbers from the table

MODEL ANSWER

T1- The table shows an inverse relationship: as female secondary enrollment rises, TFR falls.
E2- Norvale has only 22% enrollment and a TFR of 5.4, while Ruzhan has 97% enrollment and a TFR of just 1.3.
A3- This matters because the same pattern shows up around the world: places where more girls stay in school tend to have smaller families, though four countries alone can't prove that one causes the other.

Common mistake: "Education and fertility are related." It doesn't say HOW they're related and uses no numbers from the table.

Why the model works: It names the direction (inverse) and backs it up with data from two countries.

C. Explain ONE reason why increased access to education for women leads to lower fertility rates. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a mechanism: delayed marriage, paid work, family-planning knowledge, or decision-making power
- Explains how that mechanism results in fewer children

MODEL ANSWER

T1- Women who stay in school longer tend to have fewer children because they start families later and have more career options.
E2- A girl who finishes secondary school usually marries and has her first child years later, and she is more likely to work for wages.
A3- Because she starts later and each child costs her more in lost income (opportunity cost), she has fewer children overall.

Common mistake: "Educated women are smarter, so they have fewer kids." Not a real mechanism. It doesn't explain WHAT changes in her life to lower fertility.

Why the model works: It picks one mechanism (marrying later + working) and walks through how it leads to fewer births.

D. Using data from the table, explain how urbanization can contribute to declining fertility rates. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses table data showing more-urban countries have lower TFR
- Explains why: children cost more or aren't farm labor in cities, or cities offer better access to health care/contraception

MODEL ANSWER

T1- More-urban countries have lower fertility because children cost more and help less in cities.
E2- In the table, Norvale is 24% urban with a TFR of 5.4, while Tamboa is 71% urban with a TFR of 1.9.
A3- In cities, children don't provide farm labor and housing and schooling cost more, so children become an economic cost rather than an asset, which leads families to have fewer.

Common mistake: “Cities are crowded, so people have fewer kids.” No data from the table, and 'crowded' isn't the real reason. Explain costs, jobs, or health care.

Why the model works: It quotes the table AND explains the asset-to-cost shift.

E. Describe ONE likely change in the shape of Kestria's population pyramid resulting from the change in its total fertility rate between 1995 and 2020. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes a shape change: narrower base, or a wider working-age middle
- Links it to Kestria's falling TFR (5.8 → 3.1)

MODEL ANSWER

T1- Kestria's pyramid will get a narrower base because women are having fewer children.
E2- Its TFR fell from 5.8 in 1995 to 3.1 in 2020, so the youngest bars at the bottom will shrink compared with the older bars above them.
A3- Meanwhile the large group born in the 1990s moves up into the working ages, so the pyramid becomes a slimmer triangle with a bulge in the middle. If its TFR keeps falling toward replacement level (2.1), it will slowly turn into more of a beehive shape.

Common mistake: “Kestria's pyramid will change shape.” It says the shape changes but not HOW. Say which part gets narrower or wider.

Why the model works: It links the data (5.8 → 3.1) to a specific shape change (narrower base, wider middle).

F. Explain ONE economic challenge Ruzhan is likely to face as a result of its total fertility rate. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Recognizes that TFR is below replacement (≈2.1), so the population ages or shrinks
- Explains an economic challenge: labor shortage, pension/health costs, smaller tax base, dependency ratio

MODEL ANSWER

T1- Ruzhan will face an aging population with too few workers because its fertility is below replacement level.
E2- Its TFR of 1.3 is well below 2.1, so fewer young people enter the workforce each year while more people retire.
A3- As retirees grow in number compared with workers, the elderly dependency ratio rises, which can cause labor shortages and leave fewer taxpayers to pay for pensions and health care.

Common mistake: "Ruzhan's population will go down." That's a population change, not an economic challenge, and it doesn't explain the effect on jobs or money.

Why the model works: It connects low TFR → aging → a specific economic problem (labor shortages, pension costs).

G. Explain ONE limitation of using national-scale data, such as the data in the table, to understand fertility patterns. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names variation hidden by national averages (urban/rural, regional, income, ethnic/religious)
- Explains why that hides important patterns or leads to wrong conclusions

MODEL ANSWER

T1- One limitation is that national data are averages that hide differences within a country.
E2- For example, Kestria's TFR of 3.1 could combine cities near 2.0 with rural regions above 5.0.
A3- Because of this, the table can't show where fertility is actually high or why, so policies based on it could miss the places that need them most.

Common mistake: "The table doesn't have enough countries." That's about sample size, not scale. The question asks what national-level data hides.

Why the model works: It names the hidden variation (cities vs. rural areas) and explains why that leads to wrong conclusions.

Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.